

WATERMARKING BY MULTIREOLUTION HADAMARD TRANSFORM

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ABSTRACT

In this paper a watermarking algorithm for digital images is presented. The method operates in the multi-resolution Hadamard transform domain. The watermark is constructed as a normally distributed random vector and is inserted into the perceptually most significant spectral components of the image. It is found that insertion of watermark under these conditions makes the watermark robust to image processing and common geometric operations. By exploiting the statistical properties of the embedded sequence, the mark can be found without using the original image. The algorithm is benchmarked against StirMark and compared to an escrow watermarking technique.

1. INTRODUCTION

While there are many advantages of digital media and their communication, shortcomings do exist as well. The ease with which perfect digital copies are produced, creates great concern to the multimedia content providers. Copyright violation is a considerable source of financial loss. The International Intellectual Property Alliance (IIPA) estimates the annual loss of revenue in the U.S. motion picture industry due to piracy at USD 1.3 billion, and in the record and music industries at USD 1.7 billion [1]. Thus it is an urgent need to provide protection for multimedia distribution systems. Digital watermarking has been proposed as a valid solution to the problem of copyright protection for multimedia data in a networked environment. The most important characteristic of watermarking is its imperceptibility and robustness. Digital watermark is a short sequence of information, which consists of owner's identity or copyright information, added in a way that it is difficult to erase. In principle a digital watermark is an invisible mark inserted in digital media such as digital images, video and audio so that it can be detected at a later stage for evidence of copyright or generally against any illegal attempt to either reproduce or manipulate them in order to change their identity. Image watermarking techniques proposed so far can be divided into two main groups depending on the processing domain of host image that the watermark is embedded in [2]. One is the spatial domain group of techniques, according to which the intensity values of the pixels are modified [3,4]. The other is the frequency domain group, where the transform coefficients of the image are altered. The frequency domain approaches are the most successful for image watermarking. The transforms that are usually used are DCT, DFT,

DWT and other [8-11]. Obvious watermarking has been accepted as the state of the art solution for most of the applications. Many different ideas have been presented, most of them originating from Cox et al [8]. Barni et al [9] have improved the idea by providing a blind detection system.

1.1 Choice of the Transform

It has been proved that transforms with higher transform coding gain, as for example the discrete cosine and subband transforms, are suitable for watermarking when the processing noise is low [5,7]. But this is not usually the case even for the very basic applications, like compression. For low quality JPEG compression and DCT decomposition for embedding, it is easy to see that the processing noise approaches infinity for many high frequency bands, as they are bound to be completely eliminated. On the contrary, high frequency Hadamard transform coefficients have components in many DCT coefficients. So it is more likely that in such cases Hadamard transform bands survive. In SPIHT [6], it is still more likely to affect the high frequency DCT coefficients more than the high frequency Hadamard coefficients [7]. Any efficient compression method affects the high frequency bands of the transforms that are suitable for compression. The high frequency bands of Hadamard transform, however, are relatively

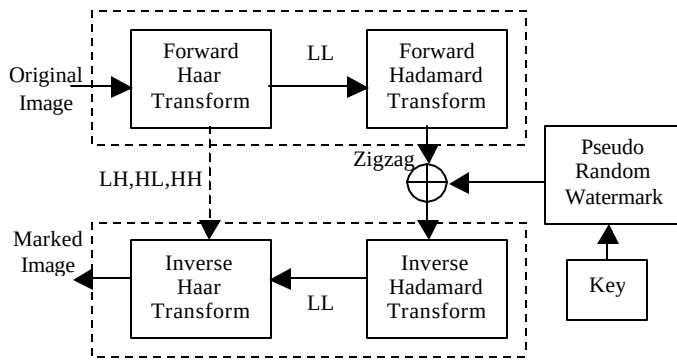


Figure 1. Watermarking strategy.

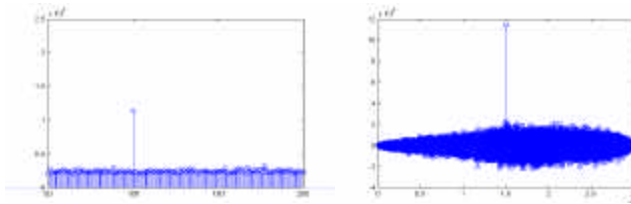


Figure 3. Response of the detector for the watermarked image 'Acropolis2'.



Figure 2. (a) Original and (b) watermarked images by means of the proposed strategy. (From top to bottom: Acropolis1, Acropolis2 and peppers).

immune to processing noise. Similarly, low quality JPEG affects the high frequency bands of subband decomposition to a much larger extent than the high frequency Hadamard bands. It is a bare fact that low frequency bands are prone to high image noise. If the high frequency bands are also affected by processing noise, it leaves us a very small room for embedding in a small number of useful mid-frequency bands. Transforms with lower transform coding gain, like the Hadamard, have many more of this 'useful mid-frequency' bands than the high coding gain transforms, at higher processing noise scenarios. Therefore, decompositions that are not efficient for compression would in

general be more immune to processing noise than decompositions with high transform coding gain [7].

2. WATERMARK STRUCTURE AND INSERTION STRATEGY

A watermark, $W=[w_1, \dots, w_n]$, is usually a random sequence of real numbers with normal distribution of zero mean and unit variance, generated by a private key. In the watermarking process, as shown in Fig.1, the image is first decomposed by means of a single level Haar transform and then only the lowest frequency band (LL) is selected. Hadamard transform is applied to this band. This double frequency transform leads to multi-resolution Hadamard frequency domain. In this domain it is possible to separate low to middle level of frequency information. This information is a prerequisite for our watermarking scheme. Hadamard transform packs most of the energy into the upper left corner of the transformed matrix. It is easy to select appropriate coefficients by zigzag scanning. We select n of those coefficients that have enough perceptual capacity to allow watermark insertion according to eq. 1 (see Appendix). Strength and length of watermark is such that give rise to robustness but also keep the unobtrusiveness criteria. Dotted arrow represents the unmodified subbands. Results shown in Fig. 2(a) represent original images (512x512), while those in Fig. 2(b) represent watermarked images.

3. WATERMARK DETECTION AND EXPERIMENTAL RESULTS

Detection is simple and straightforward. We perform the same forward multi-resolution Hadamard transform on the watermarked (and most probably processed) image, and then select the zigzag coefficients as in the embedding process, without consulting the original image. Retrieved coefficients are then cross-correlated with the watermark generated by the secret key. To verify this work, different image processing and geometric distortion operations have been applied. Since geometrical transformations typically affect the synchronization between the pseudorandom pattern of watermark and the watermarked image, the synchronization must be retrieved before the detector performs the correlation calculations. In order to detect the watermark from geometric distortions it is necessary to undo the transformations or search exhaustively throughout the image. To detect the watermark objectively (blind detection), a correlation detector is used as described in [9]. To justify our hypothesis, detector results without any attack are presented in Table 1. z is the correlation measure (eq. 2), and T_z stands for the threshold value as described in [9] (eq. 3). For double check, we also use a similarity measure (eq. 4) [8].

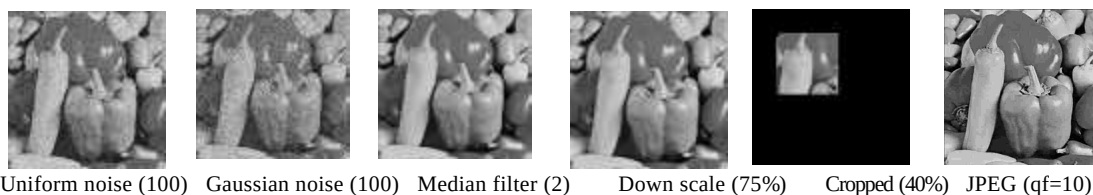


Figure 4. Examples of 'attacks' on the test image *peppers*

In all tests a watermark length of $n=15000$ was used. We modified the image by skipping the first 5000 transform coefficients and selecting the next n coefficients. A fixed strength factor α of 0.2 was used throughout.

Fig.3 shows the response of the watermark detector to 150 different keys among which only one matches the watermark present in the particular image. The positive response due to the correct watermark is dominant as compared to incorrect watermarks.

Watermarked images are subjected to different image processing operations for their robustness test, as shown in Fig.4 for the test image *peppers*. Table 2 summarizes the watermark detection results for different images and various attacks. The first column of the table represents correlation values, the second column represents threshold measures and the third column represents similarity values. It is important to mention here that all tests were performed to an extent that the image was severely degraded. All of the correlation values are above their corresponding thresholds, i.e. $z > T'_z$. Similarity measure is also well above its threshold which was set to 6, as in [9].

The proposed scheme is also tested against Stirmark [12]. If the attack is not very strong, it is possible to detect the watermark. Some of the results are presented in Table 3 for the test image *peppers*.

Robustness comparisons between the proposed scheme and that of [13], are presented in Table 4:

- **JPEG compression:** Comparison is made for detection under given minimum quality factor.
- **Scaling:** Down scaling with respect to original image size is presented. Detection is achieved by rescaling the image back to its original size for both of the schemes.
- **Cropping:** Presented values give the amount of data removed with respect to the original image during cropping process. For the detection in scheme [13], missing portion is replaced with that from the original unmarked image. For our scheme the original image is not available, so the cropped image is padded with zeros.
- **Noise distribution:** The proposed scheme is robust to random noise to an extent that the image is severely degraded.
- **Multiple watermarking:** Both schemes can detect multiple watermarks.

Table 1					Table 2					
Images	PSNR (dB)	z	$\hat{O}'_z$	Sim	Median filtering			Down scaling		
Baboon	38.29	8.52	2.89	18.29	2.0	1.6	7.2	4.1	2.5	10.2
Peppers	38.54	7.64	2.58	17.08	3.7	2.3	9.0	3.2	2.3	8.0
Pmachine	36.48	10.24	3.39	16.98	3.3	2.6	6.8	4.7	3.1	8.4
Acropolis1	36.38	12.19	3.94	17.75	6.9	3.4	11.2	5.9	3.7	9.1
Acropolis2	37.34	8.94	3.07	16.66	4.0	2.5	8.9	3.9	2.9	7.7

Table 2 (continued)											
Images	Uniform noise			Gaussian noise			Cropping			JPEG	
Baboon	6.3	4.9	8.3	5.6	5.0	7.3	1.6	1.2	8.4	7.6	2.9
Peppers	6.7	3.8	11.1	6.2	4.6	8.7	1.2	1.1	6.8	6.5	2.6
Pmachine	7.5	4.9	9.8	5.6	5.2	6.9	1.4	1.3	6.0	9.0	3.4
Acropolis1	8.9	4.3	12.6	8.7	4.8	11.4	2.6	2.2	7.2	10.3	4.0
Acropolis2	7.5	4.0	11.6	4.8	4.7	6.6	1.8	1.7	6.0	7.6	3.1

Median Filter				Linear Transform				Shearing			
Filter size	z	T_z	PSNR (dB)	Linear	z	T_z	PSNR (dB)	Shear (x:y)	z	T_z	PSNR (dB)
2x2	5.6	2.6	30.8	1.007	2.7	2.6	23.5	0:1	4.1	2.6	24.9
3x3	6.2	2.6	30.7	1.010	2.4	2.6	22.8	1:1	2.5	2.7	20.2
4x4	3.5	2.6	24.9	1.013	2.1	2.5	23.2	0:5	2.2	2.6	16.4

JPEG Compression				Row Column Removal				Aspect Ratio			
Quality	z	T _z	PSNR (dB)	Row: Col	z	T _z	PSNR (dB)	Ratio (x:y)	z	T _z	PSNR (dB)
10	4.2	2.6	31.8	5:17	6	2.6	33.1	1.2:1	6.9	2.6	33.3
15	5.2	2.6	33.4	17:5	6.4	2.6	33.8	8:1	6.8	2.6	34.5
Sharpening				Down Scaling				Table 4			
Range	z	T _z	PSNR (dB)	Scale	z	T _z	PSNR (dB)	Attack	Scheme [13]	Proposed scheme Peppers (%)	Proposed scheme goldhill (%)
3_3	16	3.2	26.4	1/2	7	2.6	38.1	JPEG Scaling Crop	10 % 56.25 % 75 %	10 6.25 83.33	10 6.25 79.82

4. CONCLUSION

A new watermarking scheme based on multi-resolution Hadamard transform is presented in this paper. The scheme is robust to most of the image processing, geometric operations and low valued StirMark manipulations. It is proved that the proposed oblivious method, compared to existing schemes, is more robust.

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APPENDIX

If x_i is a transform coefficient, x_i^* the corresponding watermarked coefficient, w_i the watermark element, a the watermark strength, and $i \in [0, n]$, with n the number of watermarked coefficients, then the embedding, the correlation measure, the threshold and the similarity formulae are respectively:

$$x_i^* = x_i + a |x_i| w_i \quad (1) \quad z = \frac{1}{n} \sum_{i=1}^n w_i x_i^* \quad (2) \quad T'_z = \frac{a}{3n} \sum_{i=1}^n |x_i^*| \quad (3) \quad \text{sim} = \frac{WX^*}{\sqrt{X^* X^*}} \quad (4)$$